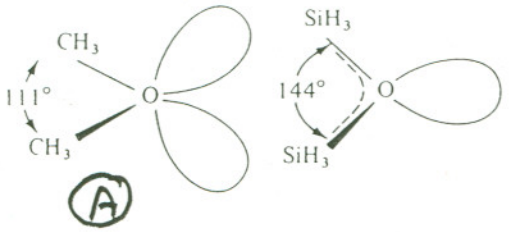
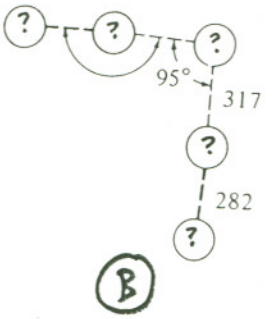
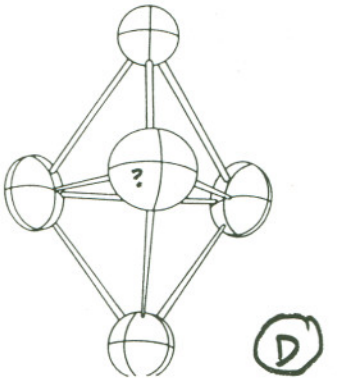
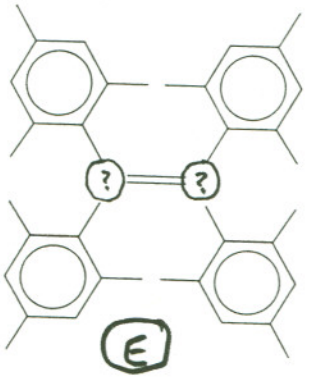
INORGANIC CUMULATIVE EXAMINATION
April 24, 1993

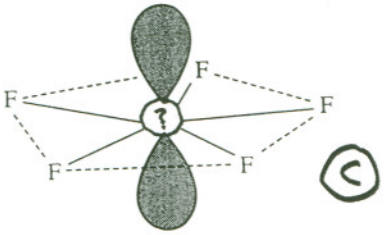
Each of the textbook figures on the attached two sheets represents a unique bit of inorganic chemistry --- a special family of compounds, a special structural story, a special bonding story, etc. Choose any 10 of those shown and give as detailed an account of the special features as possible in the time available. If the chemical symbols are not on the figure, be sure to identify what they would be.

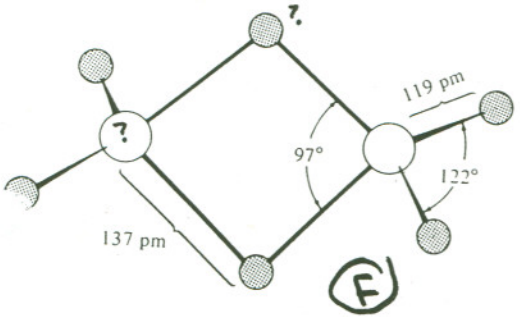

CH3
111°
CH3
O
SiH3
144°
SiH3
O
A

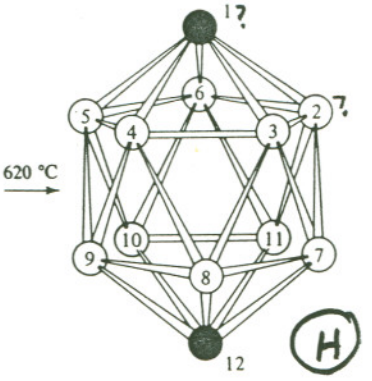

?
95°
317
282
B

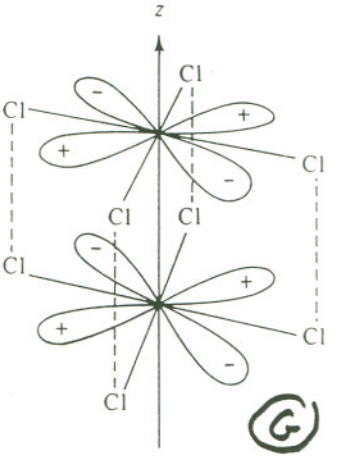

?
D

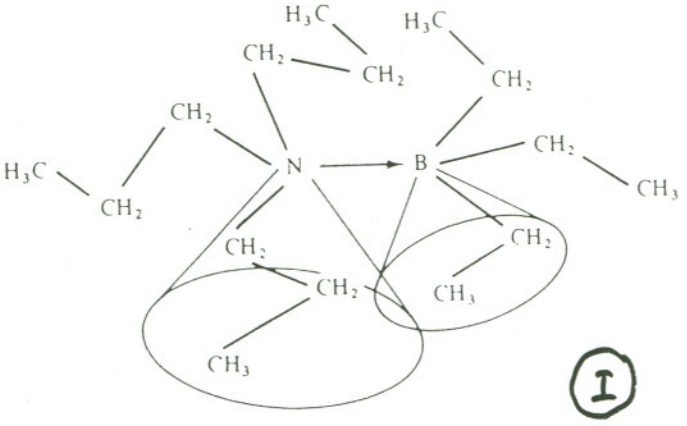

?
?
E

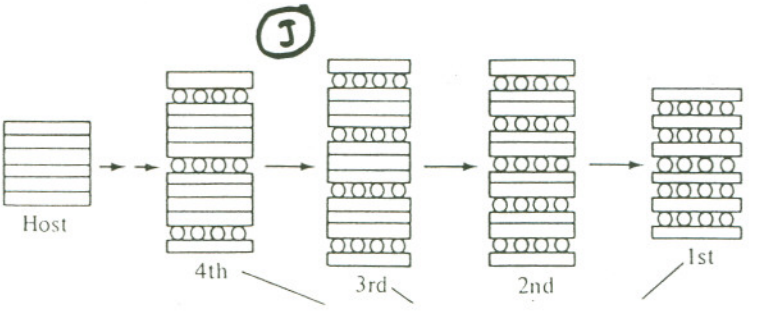

F
F
F
F
F
?
C


?
?
119 pm
97°
122°
137 pm
F

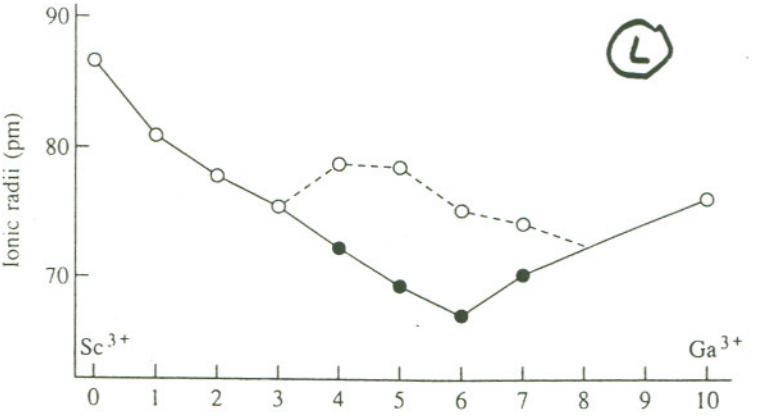

620 °C
1?
2?
3
4
5
6
7
8
9
10
11
12
H

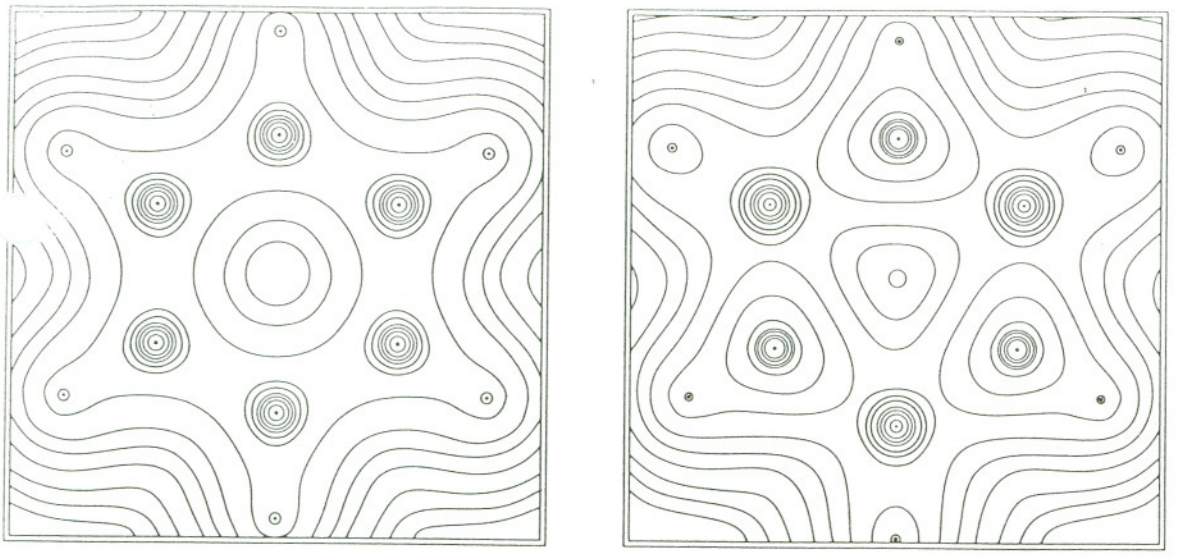

z
Cl
Cl
Cl
Cl
Cl
Cl
Cl
Cl
G


H3C
H3C
CH2
CH2
CH2
CH2
H3C
CH2
N
B
CH2
CH3
CH2
CH2
CH2
CH3
CH3
I


J
Host
4th
3rd
2nd
1st

x
z
z
z
K


L
Ionic radii (pm)
90
80
70
Sc3+
Ga3+
0
1
2
3
4
5
6
7
8
9
10



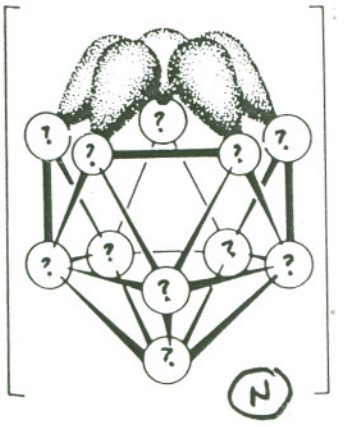
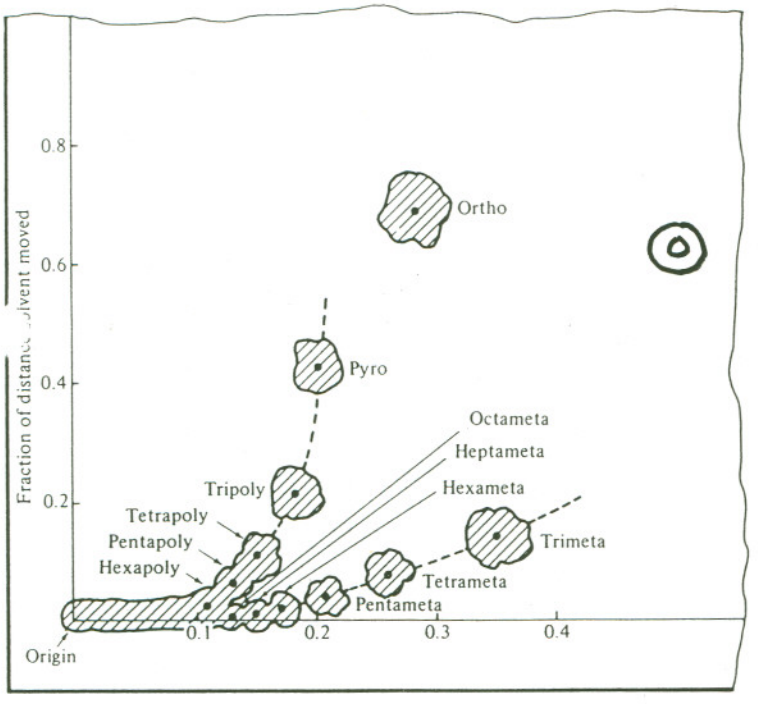
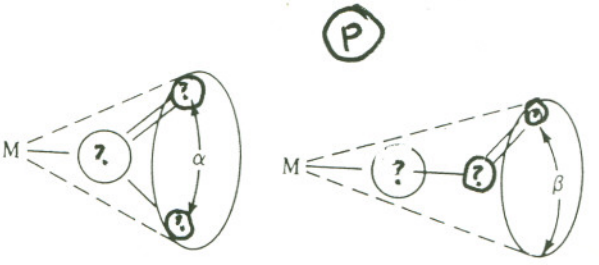
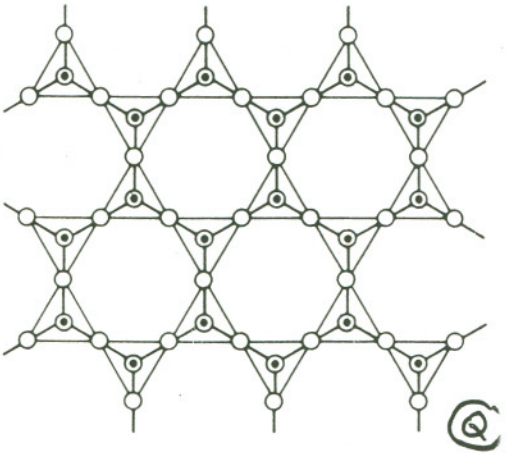
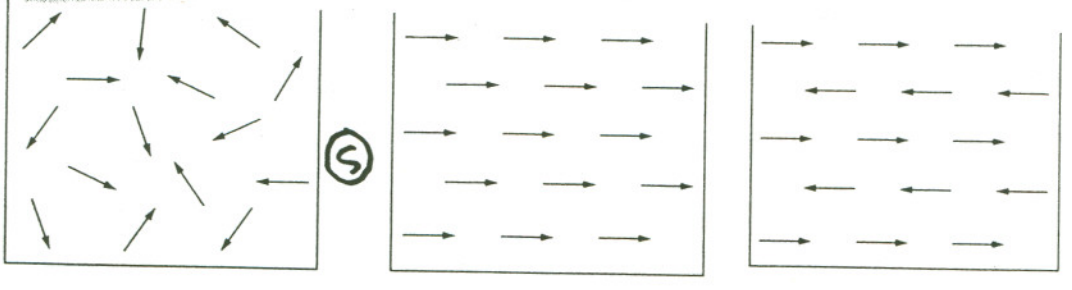
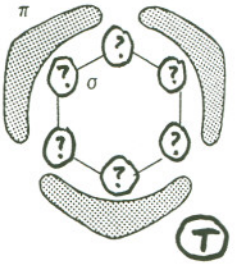


Fig. 16.22 Contour map of the charge density in the molecular plane of (a) benzene







Formula	Planar	Trigonal prismatic	Octahedral	Experimental
MA_5B	1	1	1	1
MA_4B_2	3	3	2	2
MA_3B_3	3	3	2	2

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